Java virtual machine (JVM). The JVM represents a rich signal source, exposing logs (https://openjdk.java.net/jeps/158) as well as a number of metrics, from heap memory usage to threads, and you have many tools at your disposal to monitor the JVM performance (https://sematext.com/guides/java-monitoring/).

One thing most Linux systems (bare metal and VMs) have in common these days is that they are using systemd, so let's have a look at certain aspects that are, in general, applicable to systems where users have access to the operating system. Rather than relying on textual logs, systemd comes with its own subsystem, called `journalctl`, which uses a binary format to store logs, with the advantage that the access is faster and the storage is more compact than text. However, this comes with the disadvantage that the traditional text-based tools (e.g., `tail`) don't work anymore.

The `journalctl` tool supports a rich interface, allowing for interactive as well as one-shot-style queries. For example, to show the logs of the systemd unit `example` for the past 12 hours,

```
journalctl --since "12 hours ago" -u example.service
```

A widely adopted concept in logging libraries and frameworks across programming languages is that of *log levels*, representing how severe or granular a certain message is. For example, with `journalctl` you could use `--priority=3` to limit the output to logs with a priority of `Error` (or higher priority).

We will now consider VMs as part of the infrastructure and cover specific characteristics in the context of other compute-related sources, such as Kubernetes. For all cases in which you have access to the VM—that is, besides serverless environments—it is generally true that you can view and query the system logs of the VM. Let's have a look at a concrete example and, with it, a proprietary solution for logs: consider an EC2 instance like the one that follows (note that the output is edited to fit; relevant parts are annotated):

Details about the boot process of the VM, including supported hardware

```
$ aws ec2 get-console-output --instance-id i-0c87223xxxxx --output text
i-0c87223xxxxx     [    0.000000] Linux version 5.8.0-1041-aws
(buildd@lcy01-amd64-017) (gcc (Ubuntu 10.3.0-1ubuntu1~20.10) 10.3.0,
 GNU ld (GNU Binutils for Ubuntu) 2.35.1)
 #43-Ubuntu SMP Wed Jul 14 14:49:43 UTC 2021 (Ubuntu 5.8.0-1041.43-aws 5.8.18)
[    0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-5.8.0-1041-aws
  root=PARTUUID=fd4c3fdd-01 ro console=tty1 console=ttyS0
  nvme_core.io_timeout=4294967295 panic=-1
[    0.000000] KERNEL supported cpus:
...
Welcome to Ubuntu 20.10!
[    8.645187] systemd[1]: Set hostname to <ubuntu>.
[    8.650813] systemd[1]: Initializing machine ID from KVM UUID.
[    8.655158] systemd[1]: Installed transient /etc/machine-id file.
...
[  OK  ] Finished Execute cloud user/final scripts.
[  OK  ] Reached target Cloud-init target.
```

Here, systemd kicks in and takes over the VM initialization.